

CENG381 Stochastic Processes

Random Walks on Graphs — Practice 1 (Answer Key)

Q1 Solution

a) Transition matrix P (order: C, L_1, L_2, L_3, L_4):

	C	L1	L2	L3	L4
C	[0	1/4	1/4	1/4	1/4]
L1	[1	0	0	0	0]
L2	[1	0	0	0	0]
L3	[1	0	0	0	0]
L4	[1	0	0	0	0]

Each leaf has only one neighbour (C), so from any leaf the walk always returns to C . From C , each leaf is equally likely.

b) Stationary distribution:

Graph has $|E| = 4$ edges. Degrees: $\deg(C)=4$, $\deg(L_i)=1$.

$$\pi(C) = \deg(C) / (2|E|) = 4/8 = 1/2$$

$$\pi(L_i) = \deg(L_i) / (2|E|) = 1/8 \quad \text{for } i = 1, 2, 3, 4$$

$$\text{Check: } 1/2 + 4 \times (1/8) = 1/2 + 1/2 = 1 \quad \checkmark$$

c) Irreducibility, period, convergence:

Irreducible: from C reach any L_i in 1 step; from any L_i reach C in 1 step; from L_i reach L_j via C in 2 steps.

→ All states communicate. $\checkmark$

Period of C : return paths from C : $C \rightarrow L_i \rightarrow C$ (length 2). Can C return in 1 step? $P(C,C)=0$, so no. In 3 steps? $C \rightarrow L_1 \rightarrow C \rightarrow L_1 \rightarrow C$ – that's 4 steps, not 3. $T(C) = \{2, 4, 6, \dots\} \rightarrow \text{period}(C) = 2$.

The walk does NOT converge because it is periodic (period 2). Fix: use the lazy walk $Q = (1/2)I + (1/2)P$, which adds self-loops and makes the chain aperiodic.

Q2 Solution

a) Transition matrix P for C_4 :

	0	1	2	3
0	[0	1/2	0	1/2]
1	[1/2	0	1/2	0]
2	[0	1/2	0	1/2]

CENG381 Stochastic Processes

Random Walks on Graphs — Practice 1 (Answer Key)

3 [1/2 0 1/2 0]

b) Stationary distribution and period:

All vertices have degree 2; $|E|=4$.

$$\pi(v) = 2/(2 \times 4) = 1/4 \quad \text{for all } v \text{ in } \{0,1,2,3\} \rightarrow \text{uniform.}$$

Period of state 0: The 4-cycle is bipartite with parts $\{0,2\}$ and $\{1,3\}$. From 0, the walk can only be at $\{0,2\}$ after an even number of steps. $T(0) = \{2, 4, 6, \dots\} \rightarrow \text{period}(0) = 2$.

Since the chain is periodic, it oscillates between the two parts and P^n does not converge. (E.g. starting from 0: at even times only states 0 or 2 can be visited.)

c) Lazy walk Q and aperiodicity:

$$Q = (1/2) \cdot I + (1/2) \cdot P:$$

	0	1	2	3
0	[1/2 1/4 0 1/4]			
1	[1/4 1/2 1/4 0]			
2	[0 1/4 1/2 1/4]			
3	[1/4 0 1/4 1/2]			

$$Q(0,0) = 1/2 > 0 \rightarrow t=1 \text{ in } T(0) \rightarrow \text{period}(0) \text{ in } Q = 1 \text{ (aperiodic). } \checkmark$$

$$\text{Stationary: } \pi \cdot Q = \pi \cdot [(1/2)I + (1/2)P] = (1/2)\pi + (1/2)\pi = \pi \quad \checkmark$$

$\pi = (1/4, 1/4, 1/4, 1/4)$ is also the stationary dist of Q.

Q3 Solution

a) Transition matrix, stationary distribution, period:

	A	B	C	D
A	[0 1/3 1/3 1/3]			
B	[1/3 0 1/3 1/3]			
C	[1/3 1/3 0 1/3]			
D	[1/3 1/3 1/3 0]			

All degrees = 3; $|E| = 6$.

$$\pi(v) = 3/(2 \times 6) = 1/4 \quad \text{for all } v. \quad \text{Uniform distribution.}$$

CENG381 Stochastic Processes

Random Walks on Graphs — Practice 1 (Answer Key)

Irreducible: every pair of vertices is directly connected. ✓

Period of A:

Return in 2 steps: $A \rightarrow B \rightarrow A$ (prob $1/3 \times 1/3 = 1/9 > 0$). So $t=2$ is in $T(A)$.

Return in 3 steps: $A \rightarrow B \rightarrow C \rightarrow A$ (prob $1/27 > 0$). So $t=3$ is in $T(A)$.

$\gcd(2,3) = 1 \rightarrow \text{period}(A) = 1$ (aperiodic). ✓

b) Verify $p_{\{AA\}}^{(n)} = 1/4 + (3/4) \cdot (-1/3)^n$:

$n = 1$:

Direct: $P(A,A) = 0$.

Formula: $1/4 + (3/4) \cdot (-1/3)^1 = 1/4 - 1/4 = 0$. ✓

$n = 2$:

Direct: $p_{\{AA\}}^{(2)} = P(A,B) \cdot P(B,A) + P(A,C) \cdot P(C,A) + P(A,D) \cdot P(D,A)$
 $= (1/3)(1/3) + (1/3)(1/3) + (1/3)(1/3) = 3 \times (1/9) = 1/3$.

Formula: $1/4 + (3/4) \cdot (1/9) = 1/4 + 1/12 = 3/12 + 1/12 = 4/12 = 1/3$.

✓

c) Smallest n with $|p_{\{AA\}}^{(n)} - \pi_A| < 0.01$:

$$|p_{\{AA\}}^{(n)} - 1/4| = (3/4) \cdot |-1/3|^n = (3/4) \cdot (1/3)^n$$

Solve $(3/4) \cdot (1/3)^n < 0.01$:

$$(1/3)^n < 0.01 \times (4/3) = 0.01333$$

$$n \cdot \ln(1/3) < \ln(0.01333)$$

$$n > \ln(0.01333) / \ln(1/3) = (-4.317) / (-1.099) \approx 3.93$$

Minimum $n = 4$ steps.

$$\text{Check } n=4: (3/4) \cdot (1/3)^4 = (3/4) \cdot (1/81) = 1/108 \approx 0.00926 < 0.01$$

✓

Yes, the walk converges to $\pi = (1/4, 1/4, 1/4, 1/4)$. Despite having no self-loops, K_4 is aperiodic because it has both even-length (2) and odd-length (3) cycles, making $\gcd=1$. The second eigenvalue is $\lambda_2 = -1/3$, so $|\lambda_2| = 1/3$ governs convergence.